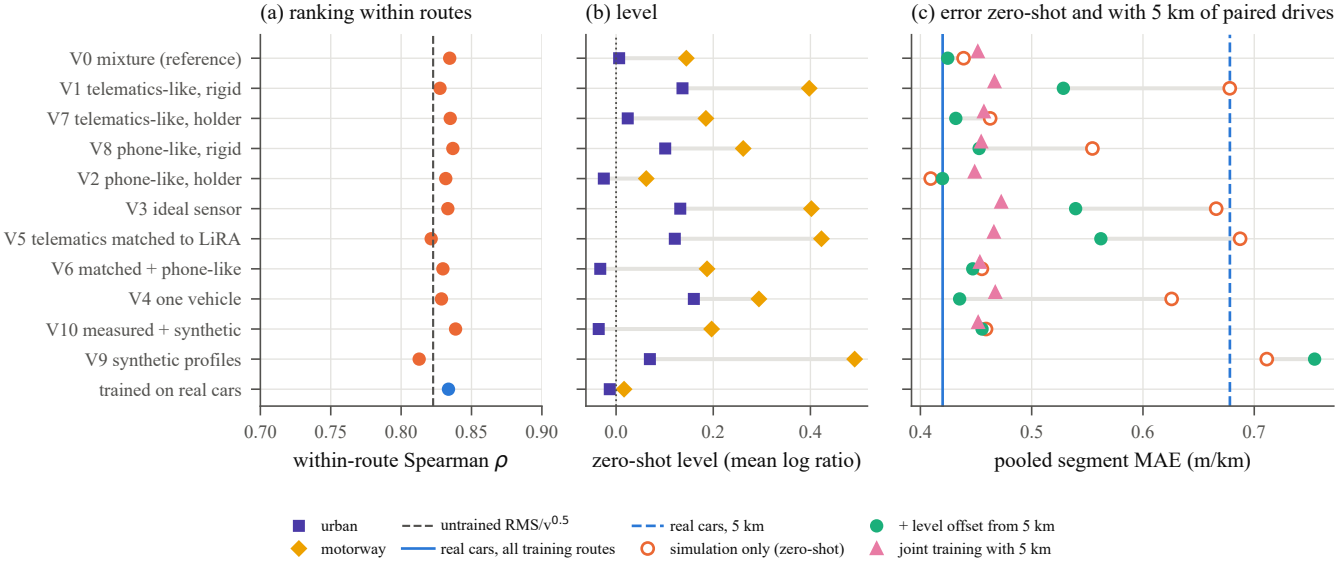
(a) ranking within routes
(b) level
(c) error zero-shot and with 5 km of paired drives
V0 mixture (reference)
V1 telematics-like, rigid
V7 telematics-like, holder
V8 phone-like, rigid
V2 phone-like, holder
V3 ideal sensor
V5 telematics matched to LiRA
V6 matched + phone-like
V4 one vehicle
V10 measured + synthetic
V9 synthetic profiles
trained on real cars
0.70
0.75
0.80
0.85
0.90
within-route Spearman ρ
0.0
0.2
0.4
zero-shot level (mean log ratio)
0.4
0.5
0.6
0.7
pooled segment MAE (m/km)
urban
motorway
untrained RMS/v^0.5
real cars, all training routes
real cars, 5 km
simulation only (zero-shot)
+ level offset from 5 km
joint training with 5 km